

OOPS

Assignment- 1

Name: Kunjal Sharma

Class: 2013

Roll no:1024150078

1. Write a program (WAP) to display "Hello World" on console display.

SOLUTION:

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World";
    return 0;
}
```

OUTPUT:

Hello World

2. Write a program to input an integer, a decimal number, and a character, and display them.

SOLUTION:

```
#include<iostream>
using namespace std;

int main() {
    int a;
    float b;
    char c;

    cout << "Input integer:" << endl;
    cin >> a;

    cout << "Input decimal:" << endl;
    cin >> b;

    cout << "Input character:" << endl;
    cin >> c;

    cout << "Integer:" << a << endl
        << "Decimal:" << b << endl
        << "Character:" << c;
}
```

OUTPUT:

Input integer:
5
Input decimal:
0.5
Input character:
K
Integer:5
Decimal:0.5
Character:K

3. Write a program to add, subtract, multiply, and divide two numbers.

SOLUTION:

```
#include <iostream>

using namespace std;

int main() {

    int a, b;

    cout << "Enter two numbers: " << endl;

    cin >> a >> b;

    int add = a+b;

    int subtract = a-b;

    int multiply = a*b;

    float divide = (float)a / b;

    cout << "Add: " << add << endl;

    cout << "Subtract: " << subtract << endl;

    cout << "Multiply: " << multiply << endl;

    cout << "Divide: " << divide << endl;

    return 0;

}
```

OUTPUT:

Enter two numbers:

10 5

Add: 15

Subtract: 5

Multiply: 50

Divide: 2

4. Write a C++ program that asks for a temperature in Celsius and displays it in Fahrenheit.
(Formula: $F = (9/5 \times C) + 32$).

SOLUTION:

```
#include <iostream>
using namespace std;
int main() {
    float c, f;
    cout << "Enter temperature in Celsius: "<<endl;
    cin >> c;
    f = (9.0 / 5) * c + 32;
    cout << "Temperature in Fahrenheit: " << f << endl;
    return 0;
}
```

OUTPUT:

Enter temperature in Celsius:

25

Temperature in Fahrenheit: 77

5. Write a program to format console output using `\n`, `\t`, `\b`, `endl`, and `setw` within `cout` statements.

SOLUTION:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    cout << "A\nB" << endl;
    cout << "A\tB" << endl;
    cout << "AB\bC" << endl;
    cout << setw(10) << "ab";
    return 0;
}
```

OUTPUT:

A

B

A B

AC

ab

6. Write a program to utilize assignment operators `+=` and `-=`.

SOLUTION:

```
#include <iostream>
using namespace std;
int main() {
    int a = 14;
    a += 5;
    cout << a << endl;
    a -= 7;
    cout << a;
    return 0;
}
```

OUTPUT:

19

12

7. Write a program to swap two numbers using a bitwise operator.

SOLUTION:

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Integers a and b are: "<<endl;
    cin >> a >> b;
    a = a ^ b;
    b = a ^ b;
    a = a ^ b;
    cout << "a: " << a << endl << "b: " << b;
    return 0;
}
```

OUTPUT:

Integers a and b are:

7 10

a: 10

b: 7

8. Write a program to calculate the library fine for late return of a book using the following rules:

- For the first 5 days, the fine is 50 paise per day.
 - For 6–10 days, the fine is 1 rupee per day.
 - For more than 10 days, the fine is 5 rupees per day.
 - If the book is returned after 30 days, the membership is cancelled.
- The program should accept the number of late days and display the fine or an appropriate message.

SOLUTION:

```
#include <iostream>
using namespace std;
int main() {
    int days;
    float fine = 0;
    cout << "Enter number of late days: ";
    cin >> days;
    if (days <= 5) {
        fine = days * 0.50;
        cout << "Fine: Rs " << fine;
    }
}
```

```

    }
    else if (days <= 10) {
        fine = (5 * 0.50) + (days - 5) * 1;
        cout << "Fine: Rs " << fine;
    }
    else if (days <= 30) {
        fine = (5 * 0.50) + (5 * 1) + (days - 10) * 5;
        cout << "Fine: Rs " << fine;
    }
    else {
        cout << "Membership cancelled due to late return.";
    }
    return 0;
}

```

OUTPUT:

Enter number of late days:

4

Fine: Rs 2

9. Write a program to perform arithmetic operations using a switch-case statement.

SOLUTION:

```

#include <iostream>
using namespace std;

int main() {
    int a, b;
    char ch;

    cout << "Enter two numbers and the operation (+, -, *, /): " << endl;
    cin >> a >> b >> ch;

    switch (ch) {
        case '+':
            cout << a + b;
            break;

        case '-':
            cout << a - b;
            break;

        case '*':
            cout << a * b;
            break;

        case '/':

```

```

        cout << (float)a / b;
        break;

    default:
        cout << "Invalid operation";
    }

    return 0;
}

```

OUTPUT:

Enter two numbers and the operation (+, -, *, /):

10 5 +

15

10. Write a program to check whether a given number is even or odd using:

(a) if-else statement

(b) conditional
operator

SOLUTION:

(a)

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n;

    cout << "Enter number:" << endl;
    cin >> n;

    if (n % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";

    return 0;
}

```

(b)

```

#include <iostream>
using namespace std;

```

```

int main() {
    int n;

    cout << "Enter number: " << endl;

```

```
    cin >> n;

    cout << (n % 2 == 0 ? "Even" : "Odd");

    return 0;
}
```

OUTPUT:

(a) Enter number:

7

Odd

(b) Enter number:

4

Even